

Q-2. Configure OSPF on each router in your topology so that Area 0 acts as the backbone and Area 1 connects through one of the routers; verify OSPF neighbor relationships using the 'show ip ospf neighbor' command. Topology



Router 0

```
Router(config)#router ospf 10
Router(config-router)#network 10.0.0.0 0.255.255.255 area 0
Router(config-router)#network 20.0.0.0 0.255.255.255 area 0
Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
00:06:30: %OSPF-5-ADJCHG: Process 10, Nbr 30.0.0.1 on Serial0/0 from LOADING to FULL, Loading Done

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
30.0.0.1         0     FULL/  -        00:00:38    20.0.0.2     Serial0/0
Router#
```

Router 1

```
Router(config)#router ospf 10
Router(config-router)#
Router(config-router)#network 20.0.0.0 0.255.255.255 area 0
Router(config-router)#network 30.0.0.0 0.255.255.255 area 0
00:06:21: %OSPF-5-ADJCHG: Process 10, Nbr 20.0.0.1 on Serial0/1 from LOADING to FULL, Loading Done

Router(config-router)#network 30.0.0.0 0.255.255.255 area 0
Router(config-router)#no network 30.0.0.0 0.255.255.255 area 0
Router(config-router)#network 30.0.0.0 0.255.255.255 area 1
Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
00:07:34: %OSPF-5-ADJCHG: Process 10, Nbr 40.0.0.1 on Serial0/0 from LOADING to FULL, Loading Done

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
20.0.0.1         0     FULL/  -        00:00:34    20.0.0.1     Serial0/1
40.0.0.1         0     FULL/  -        00:00:33    30.0.0.2     Serial0/0
Router#
```

Router 2

```
Router(config)#router ospf 10
Router(config-router)#network 30.0.0.0 0.255.255.255 area 1
Router(config-router)#
00:07:22: %OSPF-5-ADJCHG: Process 10, Nbr 30.0.0.1 on Serial0/0 from LOADING to FULL, Loading Done

Router(config-router)#network 40.0.0.0 0.255.255.255 area 1
Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
30.0.0.1	0	FULL/ -	00:00:39	30.0.0.1	Serial0/0

```
Router#
```